CPU Scheduling Algorithms - C Programs

1. FCFS Scheduling

```
#include <stdio.h>

int main() {
    int n, i;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    int bt[n], wt[n], tat[n];

    for(i=0;i<n;i++) {
        printf("Enter Burst Time for P%d: ", i+1);
        scanf("%d", &bt[i]);
    }

    wt[0] = 0;

    for(i=1;i<n;i++)
        wt[i] = wt[i-1] + bt[i-1];

    for(i=0;i<n;i++)
        tat[i] = wt[i] + bt[i];

    printf("\nProcess\tBT\tWT\tTAT\n");

    for(i=0;i<n;i++)
        printf("P%d\t%d\t%d\t%d\n", i+1, bt[i], wt[i], tat[i]);

    return 0;
}
```

Sample Input / Output

```
Sample Input:
Enter number of processes: 3
Enter Burst Time for P1: 5
Enter Burst Time for P2: 3
Enter Burst Time for P3: 8

Sample Output:
Process BT WT TAT
P1      5  0  5
P2      3  5  8
P3      8  8 16
```

2. SJF Scheduling

```
#include <stdio.h>

int main() {
    int n, i, j, temp;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    int bt[n], wt[n], tat[n];

    for(i=0;i<n;i++) {
        printf("Enter Burst Time for P%d: ", i+1);
        scanf("%d", &bt[i]);
    }

    for(i=0;i<n;i++) {
        for(j=i+1;j<n;j++) {
            if(bt[i] > bt[j]) {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    wt[0] = 0;
    for(i=1;i<n;i++)
```

```
        wt[i] = wt[i-1] + bt[i-1];

    for(i=0;i<n;i++)
        tat[i] = wt[i] + bt[i];

    printf("\nProcess\tBT\tWT\tTAT\n");

    for(i=0;i<n;i++)
        printf("P%d\t%d\t%d\t%d\n", i+1, bt[i], wt[i], tat[i]);

    return 0;
}
```

Sample Input / Output

```
Sample Input:
Enter number of processes: 3
Enter Burst Time for P1: 6
Enter Burst Time for P2: 2
Enter Burst Time for P3: 4

Sample Output:
Process BT  WT  TAT
P1      2   0   2
P2      4   2   6
P3      6   6  12
```

3. Round Robin Scheduling

```
#include <stdio.h>

int main() {
    int n, tq, i, remain;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    int bt[n], rt[n], wt[n], tat[n];

    for(i=0;i<n;i++) {
        printf("Enter Burst Time for P%d: ", i+1);
        scanf("%d", &bt[i]);
        rt[i] = bt[i];
    }

    printf("Enter Time Quantum: ");
    scanf("%d", &tq);

    remain = n;
    int time = 0;

    while(remain != 0) {
        for(i=0;i<n;i++) {
            if(rt[i] > 0) {
                if(rt[i] <= tq) {
                    time += rt[i];
                    rt[i] = 0;
                    tat[i] = time;
                    wt[i] = tat[i] - bt[i];
                    remain--;
                }
                else {
                    rt[i] -= tq;
                    time += tq;
                }
            }
        }
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for(i=0;i<n;i++)
        printf("P%d\t%d\t%d\t%d\n", i+1, bt[i], wt[i], tat[i]);

    return 0;
}
```

Sample Input / Output

```
Sample Input:
Enter number of processes: 3
```

```
Enter Burst Time for P1: 5
Enter Burst Time for P2: 4
Enter Burst Time for P3: 2
Enter Time Quantum: 2

Sample Output:
Process BT WT TAT
P1      5  6 11
P2      4  6 10
P3      2  4  6
```

4. Priority Scheduling

```
#include <stdio.h>

int main() {
    int n, i, j, temp;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    int bt[n], pr[n], wt[n], tat[n];

    for(i=0;i<n;i++) {
        printf("Enter Burst Time and Priority for P%d: ", i+1);
        scanf("%d%d", &bt[i], &pr[i]);
    }

    for(i=0;i<n;i++) {
        for(j=i+1;j<n;j++) {
            if(pr[i] > pr[j]) {
                temp = pr[i];
                pr[i] = pr[j];
                pr[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    wt[0] = 0;

    for(i=1;i<n;i++)
        wt[i] = wt[i-1] + bt[i-1];

    for(i=0;i<n;i++)
        tat[i] = wt[i] + bt[i];

    printf("\nProcess\tBT\tPriority\tWT\tTAT\n");

    for(i=0;i<n;i++)
        printf("P%d\t%d\t%d\t\t%d\t%d\n", i+1, bt[i], pr[i], wt[i], tat[i]);

    return 0;
}
```

Sample Input / Output

```
Sample Input:
Enter number of processes: 3
Enter Burst Time and Priority for P1: 5 2
Enter Burst Time and Priority for P2: 3 1
Enter Burst Time and Priority for P3: 4 3

Sample Output:
Process BT Priority WT TAT
P1      3    1     0   3
P2      5    2     3   8
P3      4    3     8  12
```